03 RESTEASE PILLOW

5 Weeks 1

Redesign of an existing manual orange juicer into a motorized, user-friendly system.

Context

Patients recovering from retinal detachment surgery are required to maintain a face-down position for extended periods to support healing. Existing face-down pillows can cause neck, shoulder and back pain, breathing difficulty, and discomfort during prolonged use. The project focused on improving ergonomic support, airflow and overall comfort during recovery.

"Comfort through every moment of recovery."

Reasearch

- Conducted **user research to understand the challenges faced during prolonged prone positioning**
- Created an **Empathy Map to identify the user's physical and emotional difficulties**
- Studied **existing face-down pillows** and identified **gaps in ergonomics, hand support and airflow.**
- Observed **user posture** and hand positioning during face-down rest.
- Conducted **prototype testing with a retinal-detachment surgery patient** to evaluate comfort and positioning.

01 Neck and shoulder strain

Prolonged prone positioning creates significant muscular strain, making extended use uncomfortable.

02 Hands need dedicated support

Users lack a comfortable place to position their hands, adding to upper-body discomfort.

03 Airflow is essential

Restricted air circulation can make the user feel suffocated and uncomfortable while resting face-down.

04 Soft forehead support is needed

Hard or uncomfortable contact around the forehead becomes problematic during long periods of use.

05 Adjustability improves comfort

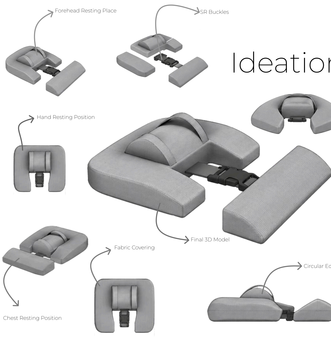
Users benefit from being able to adjust the forehead-to-shoulder distance according to their body proportions

06 Small positional adjustments matter

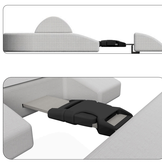
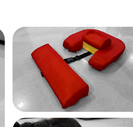
A wider, curved forehead support allows users to slightly reposition themselves without completely leaving the pillow.

Mood Board

Ideation



Renders



Prototype